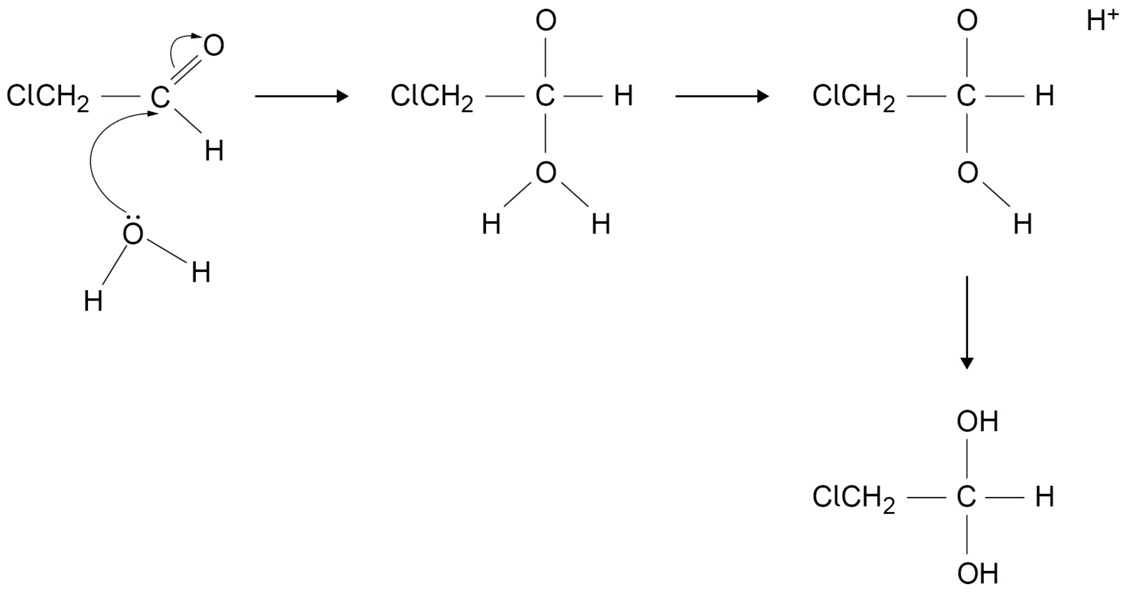
QQ8 Carbonyls

AQA 2020, AQA 2022, OCR 2022, EDEXCEL (IAL) 2023

IGC HK Exam

0 9 . 5 **Figure 6** shows an incomplete nucleophilic addition mechanism for the reaction of water with chloroethanal.

Figure 6



Complete the mechanism in **Figure 6** by adding **two** curly arrows, all relevant charges and any lone pairs of electrons involved.

[3 marks]

0 9 . 6 When an excess of water is added to ethanal a similar nucleophilic addition reaction occurs.

$$CH_3CHO(aq) + H_2O(l) \rightleftharpoons CH_3CH(OH)_2(aq)$$

Suggest why this reaction is slower than the reaction in Question **09.5**.

[3 marks]

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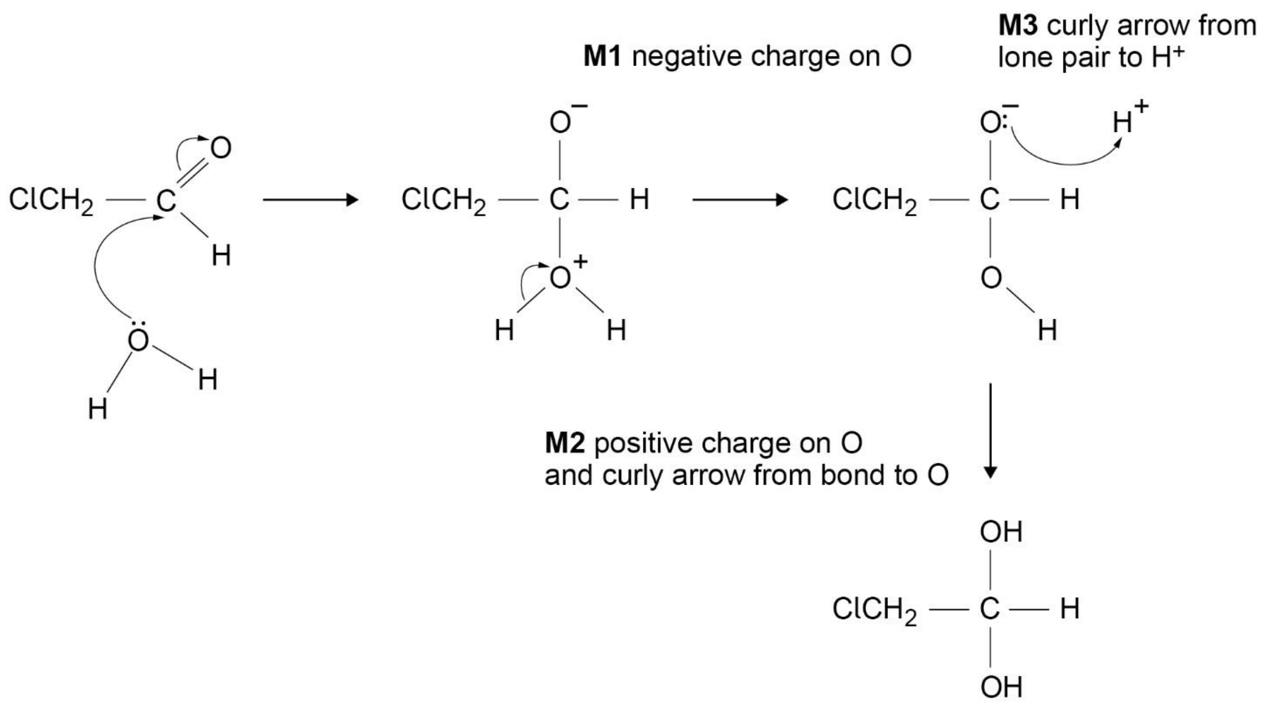
18

END OF QUESTIONS

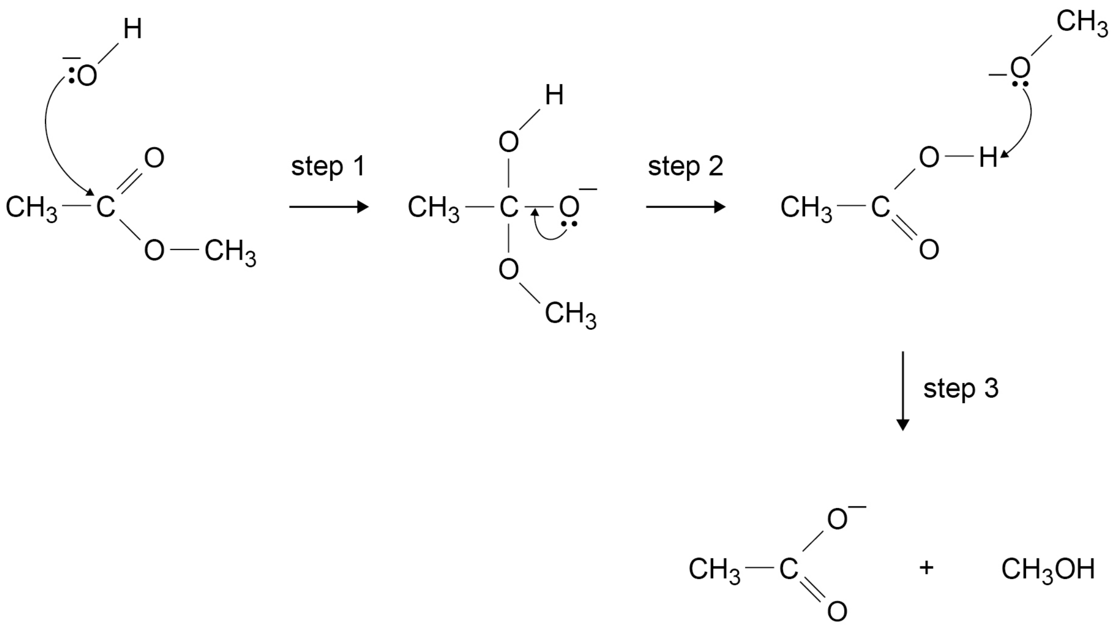
<table>
<tr><td>09.5</td><td>
</td><td>3</td></tr>
</table>

09.6	**M1** C in C=O is less δ+ / less electron deficient **M2** Because CH_3 attached is electron donating Or CH_3 has a (positive) inductive effect **M3** So higher E_a	Allow converse Ignore discussion in terms of C-Cl bond polarity Allow for **M3** water less attracted to δ+C / electron deficient C / C in C=O (so lower collision frequency/ fewer collisions with correct orientation)	1 1 1

0 7 This question is about esters.

Figure 4 shows an incomplete mechanism for the reaction of an ester with aqueous sodium hydroxide.

Figure 4



0 7 . 1 Add **three** curly arrows to complete the mechanism in **Figure 4**.

[3 marks]

0 7 . 2 Name the type of reaction shown in **Figure 4**.

[1 mark]

____

0 7 . 3 Deduce the role of the CH_3O^- ion in step **3** shown in **Figure 4**.

[1 mark]

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0 7 . 4 A triester in vegetable oil reacts with sodium hydroxide in a similar way.

Give a use for a product of this reaction.

[1 mark]

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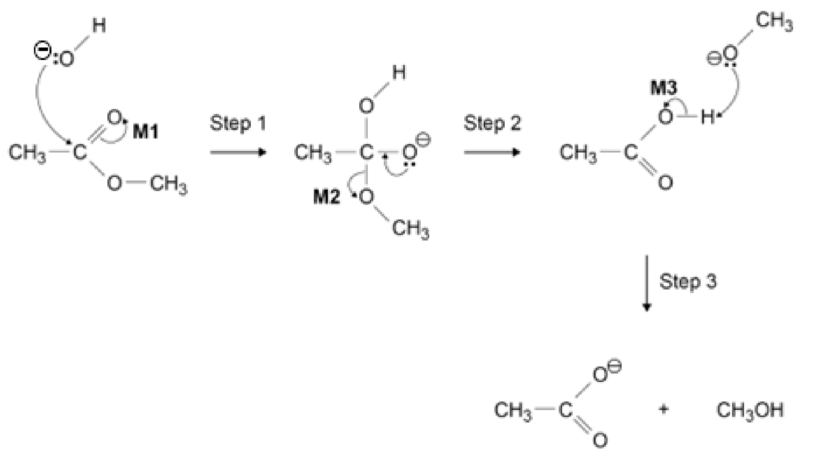
6

Question	Answers	Additional Comments/Guidelines	Mark
07.1		M1: Arrow from C=O bond to O M2 Arrow from correct C-O bond to O M3 Arrow from O-H bond to O	3 (3 x AO3)

Question	Answers	Additional Comments/Guidelines	Mark
07.2	(Alkaline/base) hydrolysis		1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
07.3	Base	Allow proton acceptor Ignore ref to Bronsted Lowry	1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
07.4	Soap only		1 (AO1)

18 This question is about carbonyl compounds.

(a) **(i)** Describe a chemical test to confirm the presence of a carbonyl group.

How could the product of this test be used to identify the carbonyl compound?

Your answer should **not** include spectroscopy.

....

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....

....

.... **[3]**

(ii) Describe a chemical test that would identify whether a carbonyl compound is an aldehyde.

Your answer **should** include the reagent and observations.

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.... **[1]**

Question			Answer	Marks	AO element	Guidance
18	(a)	(i)	(Add) 2,4-dinitrophenylhydrazine **AND** orange/yellow/red precipitate ✓ Take melting point (of crystals) ✓ Compare to known values/database ✓	3	AO1.2 × 3	**ALLOW** errors in spelling **ALLOW** 2,4(-)DNP **OR** 2,4(-)DNPH **ALLOW** Brady's reagent or Brady's Test **ALLOW** solid **OR** crystals **OR** ppt as alternatives for precipitate **Mark second and third points independently of response for first marking point** **DO NOT ALLOW** 2nd and 3rd marks for taking and comparing boiling points **OR** chromatograms
18	(a)	(ii)	Tollens' (reagent) **AND** Silver (mirror/precipitate/ppt/solid) ✓	1	AO1.2	**ALLOW** ammoniacal silver nitrate **OR** Ag^+/NH_3 **ALLOW** black ppt **OR** grey ppt **ALLOW** $Cr_2O_7^{2-}/H^+$ **AND** Turns green ✓ **IGNORE** reference to conditions, e.g. Heat or reflux -- **IF** other reagents are seen e.g. Fehling's or Benedict's, contact your Team Leader

21 Hexane-2,5-dione, $CH_3COCH_2CH_2COCH_3$, is a toxic compound formed in the human body if hexane is consumed.

(a) Complete the table for hexane-2,5-dione.

Name the organic product formed if a reaction takes place.

(2)

Reagent and conditions	Reaction (✓ / ✗)	Name of organic product (if formed)
refluxed with excess acidified potassium dichromate(VI)		
excess lithium tetrahydridoaluminate(III) in dry ether		

(b) State the observation when hexane-2,5-dione reacts with iodine in the presence of alkali.

(1)

....

....

(c) Hexane-2,5-dione reacts with **excess** hydrogen cyanide, HCN, in the presence of potassium cyanide, KCN.

(i) Name the type and mechanism of this reaction.

(1)

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(ii) Draw the structure of the product.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(d) (i) Give the observation when 2,4-dinitrophenylhydrazine (Brady's reagent) reacts with hexane-2,5-dione.

(1)

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(ii) Describe, in outline, how the product of this reaction may be used to confirm the identity of hexane-2,5-dione. Experimental details are not required.

(2)

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(Total for Question 21 = 8 marks)

TOTAL FOR SECTION B = 50 MARKS

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>21(a)</td><td>An answer that makes reference to the following points:

• row for oxidation correct (1)

• row for reduction correct (1)</td><td>Example of an answer:
<table>
<tr><th>Reagent and conditions</th><th>Reaction (✓/✗)</th><th>Name of organic product (if formed)</th></tr>
<tr><td>refluxed with excess acidified potassium dichromate(VI)</td><td>✗</td><td>(N/A)</td></tr>
<tr><td>excess lithium tetrahydrioaluminate(III) in dry ether</td><td>✓</td><td>hexane-2,5-diol</td></tr>
</table>
Accept no product or blank in first row
Accept 2,5-hexanediol in second row
Ignore errors with spaces, commas and missing ne or e in hexane
Do not award hex-2,3-diol</td><td>(2)</td></tr>
</table>

Question Number	Answer	Additional Guidance	Mark
21(b)	An answer that makes reference to the following point: • (pale) **yellow** crystals	Allow precipitate / ppt / ppte / solid Allow antiseptic smell Ignore formulae even if incorrect Do not award yellow-orange Use the list principle: if two answers and one correct and one wrong, no credit.	**(1)**

Question Number	Answer	Additional Guidance	Mark
21(c)(i)	An answer that makes reference to the following point: • nucleophilic addition	Do not award S_N1 or S_N2	**(1)**

Question Number	Answer	Additional Guidance	Mark
21(c)(ii)	An answer that makes reference to the following point: • $CH_3C(OH)(CN)CH_2CH_2C(OH)(CN)CH_3$	Allow displayed / skeletal / any combination Do not award missing hydrogens or single bonds shown between C and N. If two structures are given both must be correct.	**(1)**

Question Number	Answer	Additional Guidance	Mark
21(d)(i)	An answer that makes reference to the following point: • orange precipitate	Allow yellow / red Allow crystals / solid / ppt / ppte Ignore modifiers e.g., dark/light/brick Do not award reddish-brown	**(1)**

Question Number	Answer	Additional Guidance	Mark
21(d)(ii)	A description that makes reference to the following points: • (re)crystallise **(1)** • measure the **melting** temperature and compare with known values **(1)**	Ignore purify the product Allow refer to database, etc. Ignore NMR / mass spec. etc.	**(2)**

(Total for Question 21 = 8 marks)
TOTAL FOR SECTION B = 50 MARKS